

Servlet Programming

Introduction:

A Servlet is a Java program used to create dynamic web applications. It runs on a web server and handles requests from clients (browsers) and sends responses. Servlets work on the server side and are part of Java web technology.

How Servlets Work:

1. The client sends a request to the server.
2. The server forwards the request to the servlet.
3. The servlet processes the request.
4. The servlet generates a response.
5. The response is sent back to the client.

Servlet Life Cycle:

1. `init()` – Initializes the servlet
2. `service()` – Handles client requests
3. `destroy()` – Destroys the servlet

Advantages of Servlets:

1. Platform independent (runs on any system with Java)
2. Faster than CGI (uses threads instead of processes)
3. Efficient and scalable
4. Secure (Java security features)
5. Reusable and maintainable code

Disadvantages of Servlets:

1. Requires knowledge of Java
2. Writing HTML in Java is difficult
3. Needs server setup (like Tomcat)
4. Debugging is harder
5. Not suitable for very small applications